

* Information Gain:

Entropy of the root node

$$\text{Gain}(S, f_1) = H(S) - \sum_{v \in \text{val } |S|} \frac{|S_v|}{|S|} H(S_v)$$

Root node $H(S) = -P_+ \log_2 P_+ - P_- \log_2 P_-$

$$H(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$H(S) \approx 0.94$$

$$H(C_1) = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8}$$

$$H(C_1) \approx 0.81$$

$$H(C_2) = 1$$

$S \rightarrow$ Entire sample from the category

$S_v \rightarrow$ After split many no. of yes or no.

$$\text{Gain}(S, f_1) = 0.94 - \left[\frac{8}{14} \times 0.81 + \frac{6}{14} \times 1 \right]$$

$$\text{Gain}(S, f_1) = 0.049$$

$$\text{Gain}(S, f_1) < \text{Gain}(S, f_2) = 0.05$$

This is how information gain is calculated.

$f_1 \quad f_2 \quad f_3 \quad \text{o/p}$

